

Take Home Lab → Day 1

Task

Go over the following topic →

Endianness, Big-Endian & Little-Endian

<https://www.geeksforgeeks.org/dsa/little-and-big-endian-mystery/>

<http://stackoverflow.com/questions/22030657/little-endian-vs-big-endian>

Question 1

```
#include <stdio.h>

void doSomething(int *x, int *y) {
    int *temp;
    temp = y;
    y = x;
    x = temp;
}

int main() {

    int a = 2025, b = 0, c = 45, d = 100;
    doSomething(&a, &b);
    if (a < c)
        doSomething(&c, &a);
    doSomething(&a, &d);

    printf("%d", a);

    return 0;
}
```

Without running the code, tell the output of the given program.

Question 2

```
int main() {
```

```

int a = 300;
char *b = (char *)&a;
*++b = 2;

printf("%d ",a);

return 0;
}

```

Without running the code, tell the output of the given program.

Question 3

Design a Calculator using function pointers in C, which takes the operation, and two values as input. You can implement basic mathematical operators (add, subtract, multiply, divide). Feel free to extend the list.

Question 4

```

#include <stdio.h>

int main() {

    unsigned int x = 3251;
    char *cp = &x;
    unsigned char c1 = *cp++, c2 = *cp;
    printf("%d\n",c1);
    printf("%d\n",c2);

    return 0;
}

```

Based on the above question discussed in the class, answer the following questions →

- What is `cp` storing, an address or a value?
- What is the size of the memory location occupied by `cp`?
- What is the size of `x`?
- What is the size of `&x`?

Question 5



As it isn't expected that you know how to pass command line arguments in C right now you can use https://www.onlinegdb.com/online_c_compiler to compile your code using command line arguments for now.

Complete the function to print the command line arguments passed to the program in reversed order.

```
#include <stdio.h>

int main(int argc, char** argv) {

    // Print in reverse order
    // .....
    // Code logic goes here
    // .....

    return 0;
}
```

Sample Input:

Operating Systems is a fun course.

Sample Output:

course.
fun
a
is
Systems
Operating